

1D: $\frac{\partial^2 f}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}$ v : vel. of wave.
 "Wave Equation"

$$f(x, t) = g(x - vt) + h(x + vt)$$

$$f(x, t) = A \cos \left(\underset{\substack{\uparrow \\ \frac{2\pi}{\lambda}}}{k} x - \underset{\substack{\downarrow \\ 2\pi\nu}}{\omega} t + \delta \right)$$

$$\underline{3D}: \quad \nabla^2 f = \frac{1}{V} \frac{\partial^2 f}{\partial t^2} \quad \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$f(x, y, z, t) = X(x) Y(y) Z(z) T(t)$$

$$\begin{array}{l} \text{divide} \\ X Y Z T \end{array} \quad Y Z T \frac{d^2 X}{dx^2} + X Z T \frac{d^2 Y}{dy^2} + X Y T \frac{d^2 Z}{dz^2} = \frac{X Y Z}{V} \frac{d^2 T}{dt^2}$$

$$\Rightarrow \underbrace{\frac{1}{X} \frac{d^2 X}{dx^2}} + \underbrace{\frac{1}{Y} \frac{d^2 Y}{dy^2}} + \underbrace{\frac{1}{Z} \frac{d^2 Z}{dz^2}} = \underbrace{\frac{1}{V} \frac{1}{T} \frac{d^2 T}{dt^2}}$$

$$\frac{1}{X} \frac{d^2 X}{dn^2} = -k_x^2 \Rightarrow \frac{d^2 X}{dn^2} + k_x^2 X = 0$$

$$\vec{r} \equiv (x, y, z) \quad \vec{k} \equiv (k_x, k_y, k_z)$$

$$k^2 = k_x^2 + k_y^2 + k_z^2$$

$$\frac{1}{Y} \frac{d^2 Y}{dy^2} = -k_y^2 \quad ; \quad \frac{1}{Z} \frac{d^2 Z}{dz^2} = -k_z^2$$

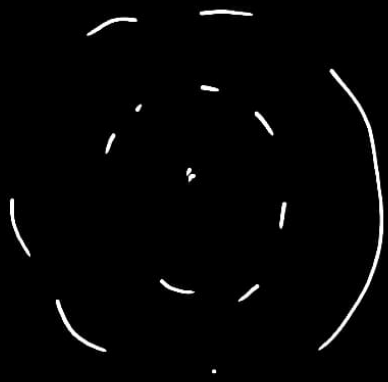
$$\omega k = \omega$$

$$-k_x^2 - k_y^2 - k_z^2 = \frac{1}{v} \frac{1}{T} \frac{d^2 T}{dt^2}$$

$$\Rightarrow \frac{d^2 T}{dt^2} = -(k_x^2 + k_y^2 + k_z^2) v^2 T = -\omega^2 T$$

$$f = \text{Re} \left[A e^{i(k_x x + k_y y + k_z z \pm \omega t + \delta)} \right]$$

$$= \text{Re} \left[A e^{i(\vec{k} \cdot \vec{r} \pm \omega t)} \right]$$



$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right)$$

$$f(r, t) = \frac{f(r, t)}{\sqrt{r^2 + y^2 + z^2}}$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right)$$

$$= \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{df}{dr} \right) = \frac{d^2 f}{dr^2} + \frac{2}{r} \frac{df}{dr}$$

$$f(r, t) = \frac{u(r, t)}{r} : \text{Ansatz}$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) = \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial}{\partial r} \left\{ \frac{u(r,t)}{r} \right\} \right]$$

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial^2 f}{\partial r^2}$$

$$f = \frac{u(r,t)}{r}$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right)$$

$$= \frac{1}{r^2} \frac{\partial^2 u}{\partial r^2}$$

$$\frac{1}{r^2} \frac{\partial^2 u}{\partial r^2}$$

$$= \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \left\{ \frac{1}{r} \frac{\partial u}{\partial r} - \frac{1}{r^2} u \right\} \right]$$

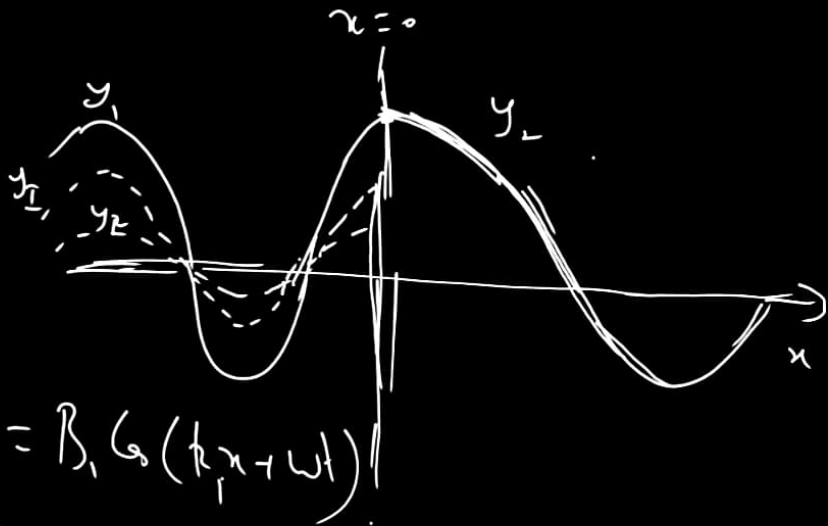
$$= \frac{1}{r^2} \frac{\partial}{\partial r} \left[r \frac{\partial u}{\partial r} - u \right]$$

$$= \frac{1}{r^2} \left\{ r \frac{\partial^2 u}{\partial r^2} + \cancel{\frac{\partial u}{\partial r}} - \cancel{\frac{\partial u}{\partial r}} \right\} = \frac{1}{r^2} \frac{\partial^2 u}{\partial r^2}$$

$$\cancel{\frac{1}{r}} \frac{\partial \tilde{u}}{\partial r} = \frac{1}{v} \cancel{\frac{1}{r}} \frac{\partial \tilde{u}}{\partial t} \Rightarrow \frac{\partial \tilde{u}}{\partial r} = \frac{1}{v} \frac{\partial \tilde{u}}{\partial t}$$

$$f(r, t) = \frac{1}{r} e^{i(\vec{k} \cdot \vec{r} \pm \omega t)} \quad u = g(r \pm vt)$$

Waves at discontinuities.



$$y_I = A_1 \cos(k_1 x - \omega t)$$

$$y_R = B_1 \cos(-k_1 x - \omega t) = B_1 \cos(k_1 x + \omega t)$$

$$y_T = A_2 \cos(k_2 x - \omega t)$$

$$\omega = v k$$

$$y_I = y_I + y_R = A_1 \cos(k_1 x - \omega t) + B_1 \cos(k_1 x + \omega t)$$

$$y_T = y_T = A_2 \cos(k_2 x - \omega t)$$

$$\text{At } x=0, \quad y_1 = y_2 \Rightarrow A_1 + B_1 = A_2 \quad \text{--- (1)}$$

$$\left. \frac{\partial y_1}{\partial x} \right|_{x=0} = \left. \frac{\partial y_2}{\partial x} \right|_{x=0} \Rightarrow \left[-A_1 k_1 \sin(k_1 x - \omega t) - B_1 k_1 \sin(k_1 x + \omega t) \right]_{x=0} = \left[-A_2 k_2 \sin(k_2 x - \omega t) \right]_{x=0}$$

$$\frac{B_1}{A_1}, \quad \frac{A_2}{A_1}$$

$$A_1 k_1 - B_1 k_1 = A_2 k_2 \quad \text{--- (2)}$$

$$\boxed{\frac{B_1}{A_1} = \frac{k_1 - k_2}{k_1 + k_2}; \quad \frac{A_2}{A_1} = \frac{2k_1}{k_1 + k_2}}$$

$$\omega = vk ; \quad v = \sqrt{\frac{T}{\mu}} \quad v \propto \frac{1}{\sqrt{\mu}} \quad \therefore k \propto \sqrt{\mu}$$

$$\frac{B_1}{A_1} = \frac{\sqrt{\mu_1} - \sqrt{\mu_2}}{\sqrt{\mu_1} + \sqrt{\mu_2}} ; \quad \frac{A_2}{A_1} = \frac{2\sqrt{\mu_1}}{\sqrt{\mu_1} + \sqrt{\mu_2}}$$

$$\mu_2 \rightarrow \infty \quad \frac{B_1}{A_1} = -1 ; \quad \frac{A_2}{A_1} = 0$$

$$B_1 = -A_1 ; \quad A_2 = 0$$

Brief Review of Electromagnetism.

$q, \vec{v}$ $\vec{E}(\vec{r}, t), \vec{B}(\vec{r}, t)$ $\swarrow$ Electromagnetic radiation.

$$\vec{F} = q \left[\vec{E} + \vec{v} \times \vec{B} \right] \leftarrow \text{"Electromagnetic force"}$$

Electrostatics & Magnetostatics. $\vec{E}(\vec{r}), \vec{B}(\vec{r})$

Benjamin Franklin

Q = charge.

quantised

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \sum_j \frac{q q_j}{|\vec{r} - \vec{r}_j|^3} (\vec{r} - \vec{r}_j) = q \vec{E}(\vec{r})$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') d^3 r' \quad \vec{\nabla} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) = - \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}$$

$$\vec{\nabla} \cdot \vec{E} = ? \quad \vec{\nabla} \times \vec{E} = ?$$

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int_V \rho dV$$

$$\oint_S \vec{F} \cdot d\vec{a} = \int_V \vec{\nabla} \cdot \vec{F} dV \leftarrow \text{Gauss' divergence theorem.}$$

$$\int_V \vec{\nabla} \cdot \vec{E} dV = \frac{1}{\epsilon_0} \int_V \rho dV \Rightarrow \boxed{\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0} \leftarrow \text{Gauss' Law}$$

$$\begin{aligned}
 \vec{E}(\vec{r}) &= - \frac{1}{4\pi\epsilon_0} \int \rho(\vec{r}') \vec{\nabla} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) d^3r' \\
 \vec{\nabla} \times \vec{E}(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \int \rho(\vec{r}') \underbrace{\vec{\nabla} \times \vec{\nabla} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right)}_{=0} d^3r' \\
 &= 0
 \end{aligned}$$

$\uparrow$
 $f(r)$

$E(\vec{r})$

$$\left(\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \quad \vec{\nabla} \times \vec{E} = 0 \right) \leftarrow \text{stationary charges.}$$

ρ : conserved.

$$I = \int \vec{J} \cdot d\vec{a}$$

$$-\frac{d}{dt} \int_V \rho dV = -\frac{dQ}{dt} = \oint_S \vec{J} \cdot d\vec{a} = \int_V \vec{\nabla} \cdot \vec{J} dV \quad \leftarrow \text{current density.}$$

$$\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t} \leftarrow \text{Eqn. of continuity}$$

$$d\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{d\vec{\ell} \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$



$\vec{B}(\vec{r})$

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}') \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} dV \quad I d\vec{\ell} \rightarrow d^3r, \quad \vec{J} dV$$

$$\vec{\nabla} \cdot \vec{B} = 0, \quad \vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

"Steady currents"

Ampere's law

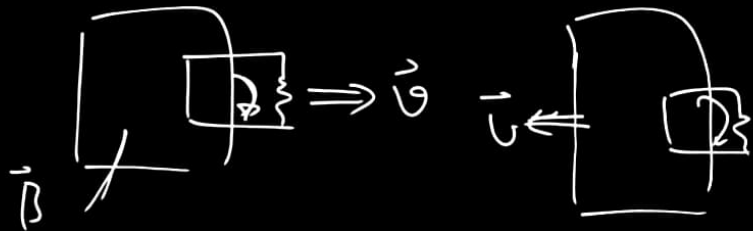
$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_{enc}$$

$$\vec{\nabla} \times \vec{E} = \frac{1}{c} \frac{\partial \vec{B}}{\partial t}; \quad \vec{\nabla} \cdot \vec{E} = \rho/\epsilon_0$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}; \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

Motional emf



$$\mathcal{E} = \oint \vec{E} \cdot d\vec{l} = - \frac{1}{dt} \int \vec{B} \cdot d\vec{a}$$

"A changing magnetic field induces an electric field"

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \vec{w}$$

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\vec{E}(\vec{r}, t)$$

$$\rho(\vec{r}, t)$$

$$0 = \vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) = \mu_0 \vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \vec{w}$$

$$\vec{\nabla} \cdot \vec{w} = -\mu_0 \frac{\partial}{\partial t} (\epsilon_0 \vec{\nabla} \cdot \vec{E}) = \mu_0 \left(-\frac{\partial \rho}{\partial t} \right) + \vec{\nabla} \cdot \vec{w}$$

$$\vec{w} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = + \mu_0 \epsilon_0 \vec{\nabla} \cdot \left(\frac{\partial \vec{E}}{\partial t} \right)$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} ; \quad \vec{\nabla} \cdot \vec{E} = \rho/\epsilon$$

$$\vec{\nabla}^2 f = \frac{1}{v} \frac{\partial^2 f}{\partial t^2}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon \frac{\partial \vec{E}}{\partial t} ; \quad \vec{\nabla} \cdot \vec{B} = 0$$

Vacuum:

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} ; \quad \vec{\nabla} \cdot \vec{E} = 0$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \epsilon \frac{\partial \vec{E}}{\partial t} ; \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$$

$$\cancel{\vec{\nabla}(\vec{\nabla} \cdot \vec{E})} - \vec{\nabla}^2 \vec{E} = -\mu_0 \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\vec{\nabla}^2 \vec{E} = \mu_0 \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\vec{\nabla}^2 \vec{B} = \mu_0 \epsilon \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$v = \frac{1}{\sqrt{\mu_0 \epsilon}} \approx 3 \times 10^8 \text{ m/s}$$

"Light is an electromagnetic wave"

$$\underline{\tilde{E}} = \underline{\tilde{E}}_0 e^{i(kz - \omega t)}$$

$$\vec{\nabla} = \frac{1}{v} \frac{\partial}{\partial t} \quad ; \quad \vec{\nabla} = \frac{1}{v} \frac{\partial}{\partial t} \quad v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$$

Light $\rightarrow$ e-m wave.

$$\vec{\nabla} f = \frac{1}{v} \frac{\partial f}{\partial t} \quad \vec{\nabla} E_n = \frac{1}{v} \frac{\partial E_n}{\partial t}$$

$$f(z, t) = A \cos(kz - \omega t + \delta) = \text{Re} \left(A e^{i(kz - \omega t + \delta)} \right)$$

$$= \text{Re} \left[A e^{i(kz - \omega t)} \right]$$

$$\vec{E}(z, t) = \vec{E}_0 e^{i(kz - \omega t)}$$

$$\vec{E}_0 = (E_{0x}, E_{0y}, E_{0z})$$

$$\vec{B}(z, t) = \vec{B}_0 e^{i(kz - \omega t)}$$

$$\vec{B}_0 = (B_{0x}, B_{0y}, B_{0z})$$

$$\vec{\nabla} \cdot \vec{E} = 0 ; \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot \vec{E} = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) \cdot \left[E_{0x} e^{i(kz - \omega t)} \hat{x} + E_{0y} e^{i(kz - \omega t)} \hat{y} + E_{0z} e^{i(kz - \omega t)} \hat{z} \right]$$

$$0 = ik E_{0z} e^{i(kz - \omega t)} \Rightarrow E_{0z} = 0$$

$\vec{k} = k\hat{z}$; $\vec{E}$ is $\perp \vec{k}$ $\leftarrow$ Electric field is transverse.

$\vec{\nabla} \cdot \vec{B} = 0 \Rightarrow B_{0z} = 0 \Rightarrow$ Magnetic field is transverse.

$\parallel^r$ to direction of propagation.

$$\begin{aligned} \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} & \vec{\nabla} \times \vec{E} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_{0x} e^{i(kz-\omega t)} & E_{0y} e^{i(kz-\omega t)} & 0 \end{vmatrix} \\ & & &= \hat{x} \left[0 - i k E_{0y} e^{i(kz-\omega t)} \right] - \hat{y} \left[0 - i k E_{0x} e^{i(kz-\omega t)} \right] + \hat{z} (0-0) \end{aligned}$$

$$\vec{\nabla} \times \vec{E} = -ik E_{0y} e^{i(kz - \omega t)} \hat{x} + ik E_{0x} e^{i(kz - \omega t)} \hat{y}$$

$$\left(-\frac{\partial \vec{B}}{\partial t} = -\frac{\partial}{\partial t} \left[\vec{B}_0 e^{i(kz - \omega t)} \right] = +i\omega \vec{B}_0 e^{i(kz - \omega t)} \right)$$

$$= i\omega \left[\hat{x} B_{0x} e^{i(kz - \omega t)} + \hat{y} B_{0y} e^{i(kz - \omega t)} \right]$$

$$-k E_{0y} = \omega B_{0x} ; \quad k E_{0x} = \omega B_{0y}$$

$$\Rightarrow B_{0x} = -\frac{k}{\omega} E_{0y} ; B_{0y} = \frac{k}{\omega} E_{0x} ; B_{0z} = 0$$

$$\vec{B}_0 = \hat{x} B_{0x} + \hat{y} B_{0y} = -\hat{x} \frac{k}{\omega} E_{0y} + \hat{y} \frac{k}{\omega} E_{0x}$$

$$\omega = ck = \frac{k}{\omega} (-\hat{x} E_{0y} + \hat{y} E_{0x})$$

$$= \frac{k}{\omega} (\hat{z} \times \vec{E}_0)$$

$$\vec{E} \perp \vec{B} \quad |\vec{B}_0| = \frac{1}{c} |\vec{E}_0|$$

$$\hat{z} \times \vec{E}_0$$

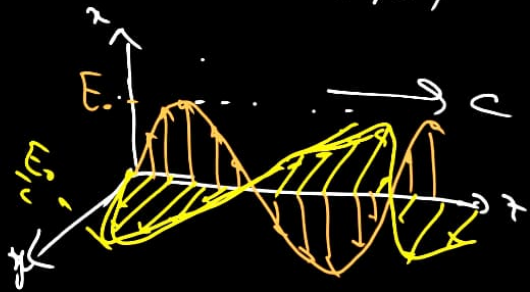
$$= \hat{z} \times (\hat{x} E_{0x} + \hat{y} E_{0y})$$

$$= \hat{y} E_{0x} - \hat{x} E_{0y}$$

$$\vec{E} = E_0 e^{i(kz - \omega t)} \hat{n} ; \quad \vec{B} = B_0 e^{i(kz - \omega t)} (-\hat{y})$$

$$\vec{E} = \text{Re}(\vec{E}) = E_0 \cos(kz - \omega t + \delta) \hat{n}$$

$$\vec{B} = \text{Re}(\vec{B}) = -E_0 \cos(kz - \omega t + \delta) \hat{y}$$



$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

Flux density $\equiv$ Energy / Area / Time.

$$\vec{g} = \frac{1}{c} \vec{S}$$



$$u_E = \frac{1}{2} \epsilon_0 \vec{E}^2 \quad \leftarrow \text{energy density}$$

$$u_B = \frac{1}{2} \frac{B^2}{\mu_0}$$

$$u = \frac{1}{2} \left[\epsilon_0 \vec{E}^2 + \frac{B^2}{\mu_0} \right]$$

$$q \rightarrow \int \rho dV$$

$$\rho \vec{v} \rightarrow \vec{J}$$

$$dW = \vec{F} \cdot d\vec{l} = q [\vec{E} + \vec{v} \times \vec{B}] \cdot \vec{v} dt = q \vec{E} \cdot \vec{v} dt$$

$$\frac{dW}{dt} = \int_V \vec{E} \cdot \vec{J} dV$$

law

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \Rightarrow \vec{J} = \frac{1}{\mu_0} (\vec{\nabla} \times \vec{B}) - \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\vec{E} \cdot \vec{J} = \frac{1}{\mu_0} \vec{E} \cdot (\vec{\nabla} \times \vec{B}) - \epsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot (\vec{E} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{B})$$

$$\begin{aligned} \Rightarrow \vec{E} \cdot (\vec{\nabla} \times \vec{B}) &= -\vec{\nabla} \cdot (\vec{E} \times \vec{B}) + \vec{B} \cdot (\vec{\nabla} \times \vec{E}) \\ &= -\vec{\nabla} \cdot (\vec{E} \times \vec{B}) - \vec{B} \cdot \frac{\partial \vec{B}}{\partial t} \end{aligned}$$

$$\begin{aligned}
 \vec{E} \cdot \vec{J} &= -\frac{1}{\mu_0} \vec{\nabla} \cdot (\vec{E} \times \vec{B}) - \frac{1}{\mu_0} \vec{B} \cdot \frac{\partial \vec{B}}{\partial t} - \epsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} \\
 &= -\frac{1}{\mu_0} \vec{\nabla} \cdot (\vec{E} \times \vec{B}) - \frac{\partial}{\partial t} \left(\frac{B^2}{2\mu_0} \right) - \frac{\partial}{\partial t} \left(\frac{\epsilon_0 E^2}{2} \right) \\
 &= -\frac{1}{\mu_0} \vec{\nabla} \cdot (\vec{E} \times \vec{B}) - \frac{\partial}{\partial t} \left[\frac{1}{2} \left(\epsilon_0 E^2 + \frac{B^2}{\mu_0} \right) \right]
 \end{aligned}$$

$$\frac{dW}{dt} : \int_V \vec{E} \cdot \vec{J} dV = -\frac{1}{\mu_0} \int_V \vec{\nabla} \cdot (\vec{E} \times \vec{B}) dV - \frac{\partial}{\partial t} \int_V \left[\frac{1}{2} \left(\epsilon_0 E^2 + \frac{B^2}{\mu_0} \right) \right] dV$$

$$\boxed{ \frac{dW}{dt} = - \oint_A \vec{S} \cdot d\vec{a} - \frac{d}{dt} \int_V u dV } \leftarrow \text{Poynting's Theorem.}$$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) \Leftarrow \text{Poynting Vector} \Rightarrow \text{Energy/Area/Time.}$$

$$\text{If } \frac{dW}{dt} = 0 \Rightarrow \oint \vec{S} \cdot d\vec{a} + \frac{d}{dt} \int_V u dV = 0 \Rightarrow \text{Energy flux density.}$$

$$\Rightarrow \int_V \vec{\nabla} \cdot \vec{S} dV + \frac{d}{dt} \int_V u dV = 0$$

$$\boxed{\frac{\partial u}{\partial t} + \vec{\nabla} \cdot \vec{S} = 0} \Leftarrow \text{Continuity eqn.}$$

Conservation of momentum:

"T"

$$\frac{d\vec{P}_{med}}{dt} = -\mu_0 \epsilon_0 \frac{\partial}{\partial t} \int_V \vec{S} dV + \text{...}$$

$$\boxed{\vec{g} = \mu_0 \epsilon_0 \vec{S} = \frac{1}{c} \vec{S}} \Leftarrow \text{Momentum Flux density}$$

$$u = \frac{1}{2} \left(\epsilon_0 E^2 + \frac{B^2}{\mu_0} \right)$$

$$= \epsilon_0 E^2$$

$$= \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)$$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

$$= \frac{c \epsilon_0^2}{\mu_0 c^2} \cos^2(kz - \omega t + \delta) \hat{z}$$

$$\vec{S} = c \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta) \hat{z} = u c \hat{z}$$

$$\vec{E} = E_0 \cos(kz - \omega t + \delta) \hat{x}$$

$$\vec{B} = \frac{E_0}{c} \cos(kz - \omega t + \delta) \hat{y}$$

$$|\vec{B}| = \frac{|\vec{E}|}{c}$$

$$\frac{B^2}{\mu_0} = \frac{E^2}{c^2 \mu_0} = \epsilon_0 E^2$$

$$\vec{g} = \frac{1}{c} \vec{S} = \frac{u}{c} \hat{z} = \frac{1}{c} u \hat{z}$$

$$\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2 ; \quad \langle \vec{S} \rangle = \frac{1}{2} c \epsilon_0 E_0^2 \hat{z}$$

$$\langle \vec{g} \rangle = \frac{1}{c} \langle u \rangle \hat{z} = \frac{1}{2c} \epsilon_0 E_0^2 \hat{z}$$

$$\text{Intensity} = \frac{\text{Power}}{\text{Area}} = \frac{\text{Energy}}{\text{Time} \cdot \text{Area}} = \langle S \rangle$$

$$I = \frac{1}{2} c \epsilon_0 E^2$$

E-M in matter.

Electric field in matter $\rightarrow$

$\nwarrow$
Bound charge density,

$$\rho_b = - \vec{\nabla} \cdot \vec{P}$$

Polarization. ($\vec{P}$)
electric dipole
moment / volume

Magnetic field in matter $\rightarrow$ Magnetisation ($\vec{M}$)

magnetic dipole moment/volume

$$\vec{J}_b = \vec{\nabla} \times \vec{M}$$

Electrodynamics $\rightarrow$ Polarization current

$$\vec{J}_p = \frac{\partial \vec{P}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{B} = 0; \quad \vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\rho = \rho_f + \rho_b = \rho_f - \vec{\nabla} \cdot \vec{P}$$

$$\vec{J} = \vec{J}_f + \vec{J}_b + \vec{J}_p = \vec{J}_f + \vec{\nabla} \times \vec{M} + \frac{\partial \vec{P}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\epsilon_0} (\rho_f - \vec{\nabla} \cdot \vec{P}) \Rightarrow \boxed{\vec{\nabla} \cdot \vec{D} = \rho_f} \quad \vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J}_f + \vec{\nabla} \times \vec{M} + \frac{\partial \vec{P}}{\partial t} \right) + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\Rightarrow \vec{\nabla} \times \left(\underbrace{\frac{\vec{B}}{\mu_0}}_{\vec{H}} - \vec{M} \right) = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

$$\Rightarrow \boxed{\vec{\nabla} \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}}$$

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} \Rightarrow \vec{B} = \mu_0 (\vec{H} + \vec{M})$$

Maxwell's Equations in matter.

$$\vec{\nabla} \cdot \vec{D} = \rho_f \quad ; \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad ; \quad \vec{\nabla} \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

$$\left\{ \begin{array}{l} \vec{D} = \epsilon_0 \vec{E} + \vec{P} \\ \vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} \end{array} \right.$$

Linear Media: $\vec{P} = \epsilon_0 \chi_e \vec{E} \Rightarrow \vec{D} = \epsilon_0 (1 + \chi_e) \vec{E}$

$$\vec{M} = \mu_0 \chi_m \vec{H} \Rightarrow \vec{B} = \mu_0 (1 + \chi_m) \vec{H} = \mu \vec{H}$$

$$\begin{array}{l} \vec{D} = \epsilon \vec{E} \\ \vec{B} = \mu \vec{H} \end{array}$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho_f}{\epsilon} ; \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} ; \quad \vec{\nabla} \times \vec{B} = \mu \vec{J}_f + \mu \epsilon \frac{\partial \vec{E}}{\partial t}$$

$$\nabla^2 \vec{E} = \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\nabla^2 \vec{B} = \mu \epsilon \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$(\rho_f = J_f = 0)$$

$$v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\mu_0 \epsilon_0} \sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}}} = \frac{c}{n}$$

refractive
index

$$n = \sqrt{\frac{\epsilon \mu}{\epsilon_0 \mu_0}}$$

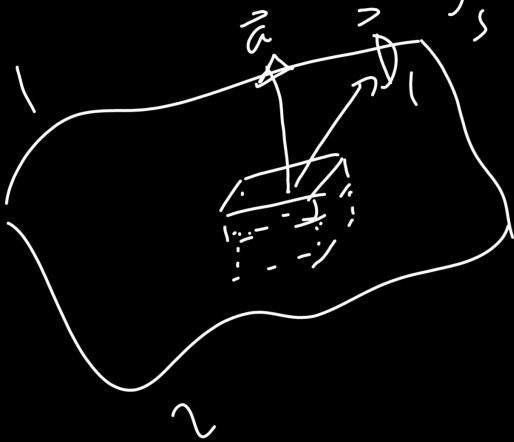
$$\mu \approx \mu_0 \quad n = \sqrt{\frac{\epsilon}{\epsilon_0}} = \sqrt{\epsilon_r} > 1$$

UCC



$$\vec{\nabla} \cdot \vec{D} = \rho_f \Rightarrow$$

$$\oint_S \vec{D} \cdot d\vec{a} = Q_f$$



$$\vec{D}_1 \cdot \vec{a} - \vec{D}_2 \cdot \vec{a} = \sigma_f a$$

$$D_1^\perp - D_2^\perp = \sigma_f$$

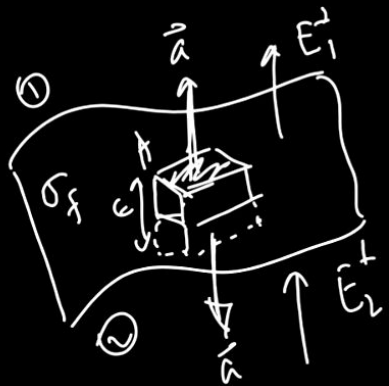
$$\vec{\nabla} \cdot \vec{D} = \rho_f \Rightarrow \oint_S \vec{D} \cdot d\vec{a} = \int_V \rho_f dV = Q_f = \sigma_f A$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow \oint_C \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{a}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \Rightarrow \oint_S \vec{B} \cdot d\vec{a} = 0$$

$$\vec{\nabla} \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t} \Rightarrow \oint_C \vec{H} \cdot d\vec{l} = I_f + \frac{\partial}{\partial t} \int_S \vec{D} \cdot d\vec{a}$$

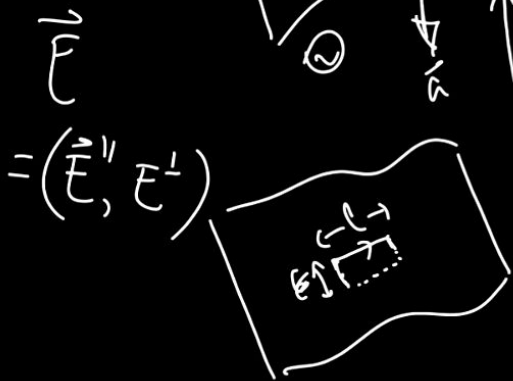
$$\vec{D} = \epsilon \vec{E}, \quad \vec{B} = \mu \vec{H}$$



$$\lim_{\epsilon \rightarrow 0} [\epsilon_1 E_1^\perp - \epsilon_2 E_2^\perp] A = \sigma_f A$$

$$\boxed{\epsilon_1 E_1^\perp - \epsilon_2 E_2^\perp = \sigma_f}$$

⊥ comp. of $\vec{E}$ discontinuous across boundary.



$$\oint_C \vec{E} \cdot d\vec{l} = - \frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{a}$$

$$\lim_{\epsilon \rightarrow 0} (\vec{E}_1 - \vec{E}_2) \cdot \vec{l} = 0 \Rightarrow \boxed{\vec{E}_1 = \vec{E}_2}$$

|| comp of $\vec{E}$ is continuous. -

$$\left. \begin{aligned}
 B_1^\perp &= B_2^\perp & ; & & \frac{\vec{B}_1''}{\mu_1} - \frac{\vec{B}_2''}{\mu_2} &= \cancel{\frac{\sigma_f}{\mu_0}} \\
 \epsilon_1 E_1^\perp - \epsilon_2 E_2^\perp &= \sigma_f & ; & & \vec{E}_1'' &= \vec{E}_2''
 \end{aligned} \right\} \text{Boundary condition.}$$

No free charges & currents + Linear medium

$$\begin{aligned}
 \epsilon_1 E_1^\perp &= \epsilon_2 E_2^\perp & ; & & \vec{E}_1'' &= \vec{E}_2'' \\
 B_1^\perp &= B_2^\perp & ; & & \frac{\vec{B}_1''}{\mu_1} &= \frac{\vec{B}_2''}{\mu_2}
 \end{aligned}$$

Incident wave at normal incidence.

$$\vec{E}_I = E_{0I} e^{i(kz - \omega t)} \hat{x}$$

$$\vec{B}_I = B_{0I} e^{i(kz - \omega t)} \hat{y}$$

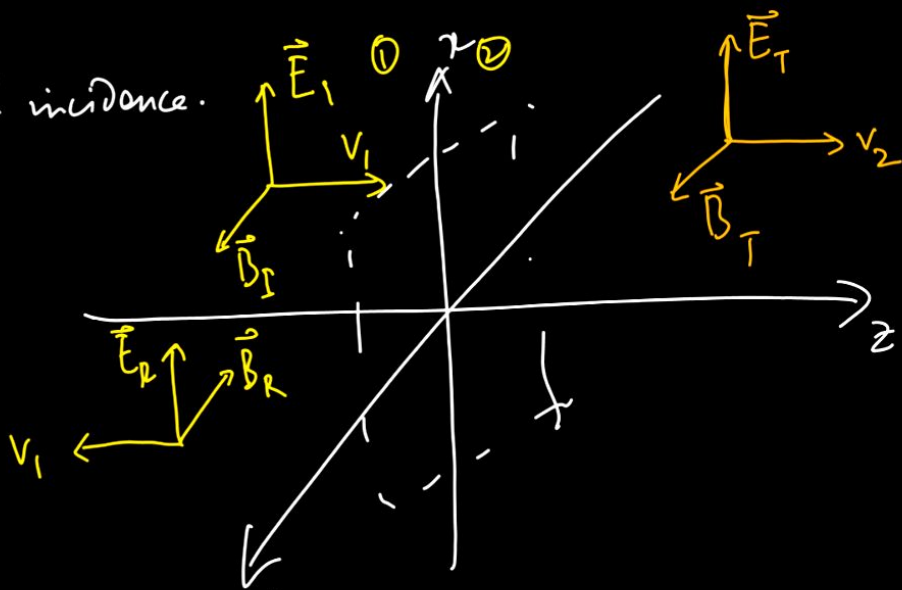
$$= \frac{E_{0I}}{v_1} e^{i(kz - \omega t)} \hat{y}$$

$$\vec{E}_R = E_{0R} e^{i(k_1 z - \omega t)} \hat{x}$$

$$\vec{B}_R = -\frac{E_{0R}}{v_1} e^{i(k_1 z - \omega t)} \hat{y}$$

$$\vec{E}_T = E_{0T} e^{i(k_2 z - \omega t)} \hat{x}$$

$$\vec{B}_T = \frac{E_{0T}}{v_2} e^{i(k_2 z - \omega t)} \hat{y}$$



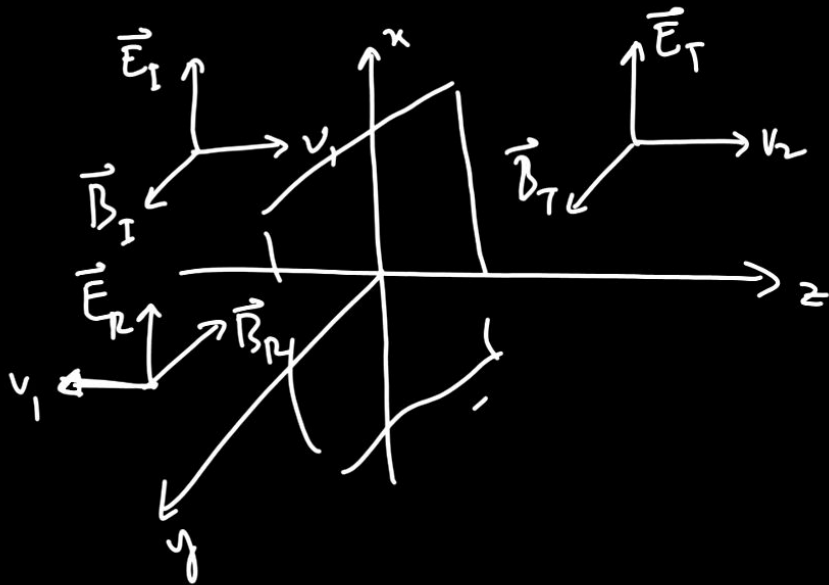
Boundary condition:

$$\epsilon_1 E_1^\perp = \epsilon_2 E_2^\perp$$

$$B_1^\perp = B_2^\perp$$

$$\vec{E}_1^\parallel = \vec{E}_2^\parallel$$

$$\frac{\vec{B}_1^\parallel}{\mu_1} = \frac{\vec{B}_2^\parallel}{\mu_2}$$



$$\vec{E}_1 = E_{01} e^{i(k_1 z - \omega t)} \hat{x}$$

$$\vec{B}_1 = \frac{E_{01}}{v_1} e^{i(k_1 z - \omega t)} \hat{y}$$

$$\vec{E}_2 = E_{02} e^{i(k_2 z - \omega t)}$$

$$\vec{B}_2 = -\frac{E_{02}}{v_2} e^{i(-k_2 z - \omega t)} \hat{y}$$

$$\vec{E}_T = E_{0T} e^{i(k_2 z - \omega t)} \hat{x}$$

$$\vec{B}_T = \frac{E_{0T}}{v_2} e^{i(k_2 z - \omega t)} \hat{y}$$

$$\checkmark E_{\sim 0I} + E_{\sim 0R} = E_{\sim 0T} \text{ at } z=0$$

$$\mu_1 \approx \mu_2 \approx \mu$$

$$v = c/n$$

$$\frac{1}{\mu_1} \left(\frac{E_{\sim 0I}}{v_1} - \frac{E_{\sim 0R}}{v_2} \right) = \left(\frac{E_{\sim 0T}}{v_2} \right) \frac{1}{\mu_2} \text{ at } z=0$$

$$n = \sqrt{\frac{\epsilon \mu}{\epsilon_0 \mu_0}} \sim \sqrt{\epsilon_r}$$

$$\Rightarrow \checkmark E_{\sim 0I} - E_{\sim 0R} = \left(\frac{\mu_1 v_1}{\mu_2 v_2} \right) E_{\sim 0T} = \beta E_{\sim 0T}$$

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2}$$

$$E_{\sim 0T} = \frac{2v_2}{v_1 + v_2} E_{\sim 0I} ; E_{\sim 0R} = \frac{v_2 - v_1}{v_2 + v_1} E_{\sim 0I}$$

$$\sim \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

$$E_{oT} = \frac{2v_L}{v_1 + v_L} E_{oI} ; \quad E_{oR} = \left(\frac{v_L - v_1}{v_1 + v_L} \right) E_{oI}$$

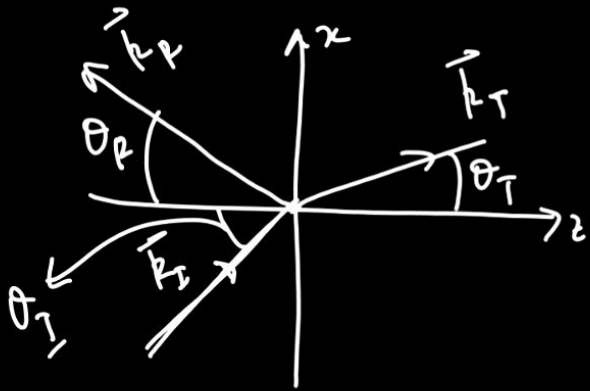
$$= \frac{n_1 - n_L}{n_1 + n_L} E_{oI}$$

$$\langle S_I \rangle = \langle S_R \rangle + \langle S_T \rangle$$

$$v_1 \epsilon_1 E_{oI}^2 = v_1 \epsilon_1 E_{oR}^2 + v_2 \epsilon_2 E_{oT}^2$$

→ Will this be true?

$$\begin{aligned} \langle S \rangle &= c \langle \mu \rangle \\ &= \frac{1}{2} c \epsilon_0 E_o^2 \\ v = \frac{1}{\sqrt{\epsilon \mu}} \Rightarrow v^2 \epsilon &= \frac{1}{\mu} \end{aligned}$$



Oblique incidence. $\vec{p} = k \hat{z}$

$$\vec{B}_I(\vec{r}, t) = \frac{1}{v_1} (\hat{k}_I \times \vec{E}_I)$$

$$\vec{B}_T(\vec{r}, t) = \frac{1}{v_2} (\hat{k}_T \times \vec{E}_T)$$

$$\vec{B}_R(\vec{r}, t) = \frac{1}{v_1} (\hat{k}_R \times \vec{E}_R)$$

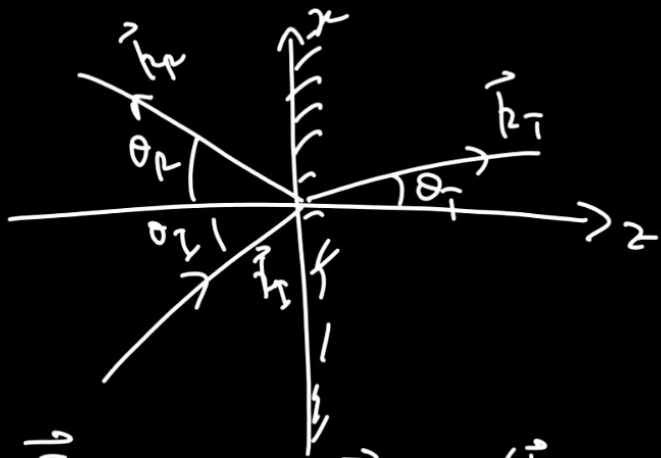
$$\vec{E}_I = E_{0I} e^{i(\vec{k}_I \cdot \vec{r} - \omega t)} \hat{n}_I$$

$$\vec{E}_R = E_{0R} e^{i(\vec{k}_R \cdot \vec{r} - \omega t)} \hat{n}_R$$

$$\vec{E}_T = E_{0T} e^{i(\vec{k}_T \cdot \vec{r} - \omega t)} \hat{n}_T$$

$$\omega = k_I v_1 = k_R v_1 = k_T v_2$$

$$k_I = k_R = k_T \frac{v_2}{v_1}$$



$$\vec{E}_I(\vec{r}, t) = \vec{E}_{0I} e^{i(\vec{k}_I \cdot \vec{r} - \omega t)}$$

$$\vec{B}_I(\vec{r}, t) = \frac{1}{c} (\hat{k}_I \times \vec{E}_I)$$

$$\vec{E}_P(\vec{r}, t) = \vec{E}_{0P} e^{i(\vec{k}_P \cdot \vec{r} - \omega t)}$$

$$\vec{B}_P(\vec{r}, t) = \frac{1}{c} (\hat{k}_P \times \vec{E}_P)$$

$$\vec{E}_T(\vec{r}, t) = \vec{E}_{0T} e^{i(\vec{k}_T \cdot \vec{r} - \omega t)}$$

$$\vec{B}_T(\vec{r}, t) = \frac{1}{c} (\hat{k}_T \times \vec{E}_T)$$

$$\vec{E}_{\sim I} + \vec{E}_{\sim R} = \vec{E}_{\sim T} \quad \text{at } z=0$$

$$\left(\right) e^{i(\vec{k}_I \cdot \vec{r} - \omega t)} + \left(\right) e^{i(\vec{k}_R \cdot \vec{r} - \omega t)}$$

$$= \left(\right) e^{i(\vec{k}_T \cdot \vec{r} - \omega t)} \quad \text{at } z=0$$

$$\vec{k}_I \cdot \vec{r} = \vec{k}_R \cdot \vec{r} = \vec{k}_T \cdot \vec{r} \quad \text{at } z=0$$

$$\Rightarrow x(k_I)_x + y(k_I)_y = x(k_R)_x + y(k_R)_y = x(k_T)_x + y(k_T)_y \quad \text{at } z=0$$

$$F_n y = 0 \Rightarrow (k_I)_n = (k_R)_n = (k_T)_n \quad \text{--- (1)}$$

$$n = 0 \Rightarrow (k_I)_y = (k_R)_y = (k_T)_y \quad \text{--- (2)}$$

Suppose $\vec{k}_I$ lies in the $n-z$ plane $\Rightarrow (k_I)_y = 0$

From (2): $(k_R)_y = (k_T)_y = 0$

So does $\vec{k}_R, \vec{k}_T$

$\rightarrow$ 1st law.

$\vec{k}_I, \vec{k}_R, \vec{k}_T$ all lie on the $n-z$ plane

$$v = \frac{c}{n}$$

$$k_I \sin \theta_I = k_R \sin \theta_R = k_T \sin \theta_T$$

$$\omega = v_I k_I = v_R k_R = v_T k_T \Rightarrow k_I = k_R = \frac{v_T}{v_I} k_T$$

$$= \frac{n_I}{n_T} k_T$$

$$\Rightarrow k_I = k_R \Rightarrow \sin \theta_I = \sin \theta_R$$

$$\Rightarrow \theta_I = \theta_R$$

$$k_I = \frac{n_I}{n_T} k_T$$

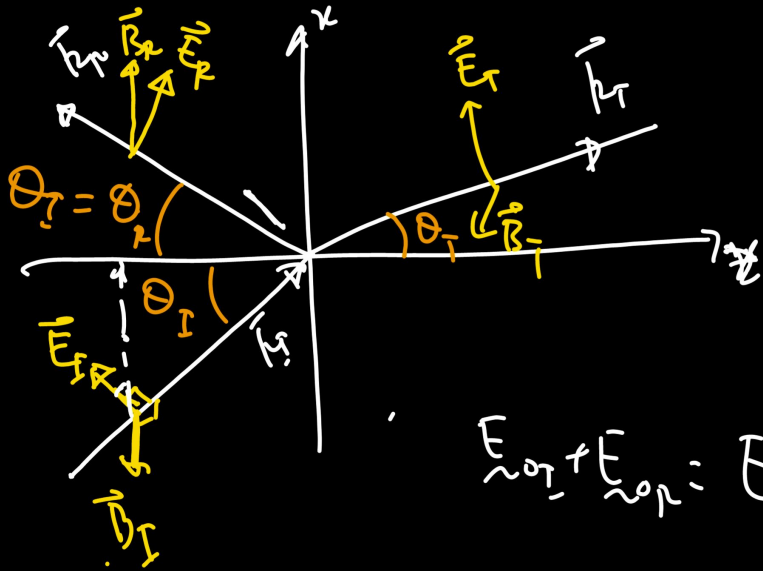
$$\Rightarrow n_I \sin \theta_I = n_T \sin \theta_T$$

Snell's law.

$$\left. \begin{aligned} \epsilon_1 E_1^\perp &= \epsilon_2 E_2^\perp \\ B_1^\perp &= B_2^\perp \\ \vec{E}_1 &= \vec{E}_2 \\ \vec{B}_1 &= \vec{B}_2 \end{aligned} \right\} \Rightarrow$$

$$\begin{aligned} \epsilon_1 \left[\vec{E}_{\sim 01} + \vec{E}_{\sim 02} \right]_z &= \epsilon_2 \left[\vec{E}_{\sim 0T} \right]_z \\ \left[\vec{B}_{\sim 01} + \vec{B}_{\sim 02} \right]_z &= \left[\vec{B}_{\sim 0T} \right]_z \\ \left[\vec{E}_{\sim 01} + \vec{E}_{\sim 02} \right]_{n,\gamma} &= \left[\vec{E}_{\sim 0T} \right]_{n,\gamma} \\ \frac{1}{\mu_1} \left(\vec{B}_{\sim 01} + \vec{B}_{\sim 02} \right)_{n,\gamma} &= \frac{1}{\mu_2} \left[\vec{B}_{\sim 0T} \right]_{n,\gamma} \end{aligned}$$

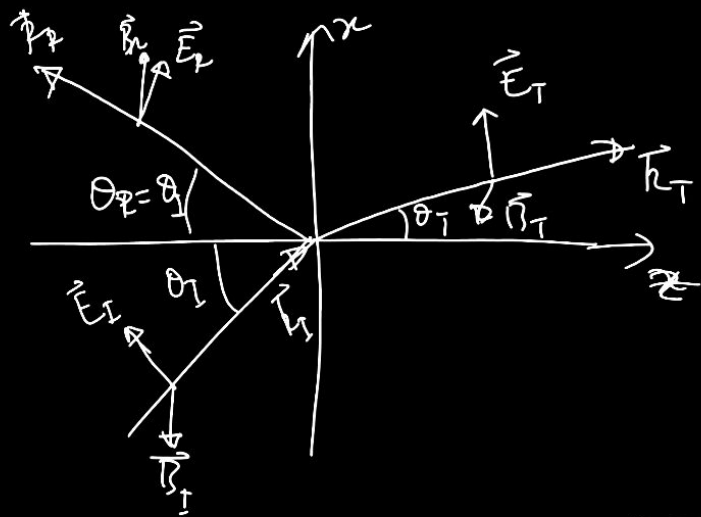
Polarisation $\parallel$ to plane of incidence



$$\epsilon_1 \left[-E_{0r} \sin \theta_I + E_{0r} \sin \theta_r \right] = \epsilon_2 \left[-E_{0T} \sin \theta_T \right]$$

$$E_{0I} \cos \theta_I + E_{0r} \cos \theta_r$$

$$E_{0r} + E_{0T} = E_{0I} \left(\frac{\cos \theta_T}{\cos \theta_I} \right) = E_{0I} \cos \theta_T$$



$$\frac{L_{\theta_7}}{L_{\theta_7 \perp}} = \alpha$$

$$E_1 \left[-E_{o1} \sin \theta_L + E_{op} \sin \theta_L \right]$$

$$E_{\sim \theta_I} G \theta_I + E_{\sim \theta_I} G \theta_I''^{\theta_I} = E_{\sim \theta_I} G \theta_I$$

$$\Rightarrow \underline{\underline{E_{\alpha_1} + E_{\alpha_2} = \alpha E_{\alpha_1}}}$$

$$\frac{1}{\mu_1} \left[\underset{\substack{= \\ E_{\sim 0I} \\ \downarrow \\ \nu_1}}{B_{\sim 0I}} - \underset{\substack{= \\ E_{\sim 0R} \\ \downarrow \\ \nu_1}}{B_{\sim 0R}} \right] = \frac{1}{\mu_2} \underset{\substack{= \\ E_{\sim 0T} \\ \downarrow \\ \nu_2}}{B_{\sim 0T}} \Rightarrow \frac{1}{\mu_1 \nu_1} \left[E_{\sim 0I} - E_{\sim 0R} \right] = \frac{1}{\mu_2 \nu_2} E_{\sim 0T}$$

$$\boxed{E_{\sim 0I} - E_{\sim 0R} = \beta E_{\sim 0T}}$$

$$\beta = \frac{\mu_1 \nu_1}{\mu_2 \nu_2} \approx \frac{\nu_1}{\nu_2} = \frac{n_2}{n_1}$$

$$\alpha = \frac{G_{\sim 0T}}{G_{\sim 0I}} = \frac{\sqrt{1 - \sin^2 \theta_T}}{\sin \theta_I}$$

$$\boxed{\begin{aligned} E_{\sim 0T} &= \frac{2}{\alpha + \beta} E_{\sim 0I} \\ E_{\sim 0R} &= \frac{\alpha - \beta}{\alpha + \beta} E_{\sim 0I} \end{aligned}} \Rightarrow \text{Fresnel Equations.}$$

$$\alpha = \frac{\cos \theta_T}{\cos \theta_I} = \frac{\sqrt{1 - \sin^2 \theta_T}}{\cos \theta_I} = \frac{\sqrt{1 - \frac{n_1^2}{n_2^2} \sin^2 \theta_I}}{\cos \theta_I}$$

$$\begin{aligned} n_1 \sin \theta_I &= n_2 \sin \theta_T \\ &= n_2 \sin \theta_T \end{aligned}$$

$$\theta_I = 0 \Rightarrow \alpha = 1$$

Grazing incidence, $\theta_I = 90^\circ$, $\alpha \rightarrow \text{diverge. ? ?}$

In conductors : $\vec{J}_f = \sigma \vec{E}$, $\vec{D} = \epsilon \vec{E}$, $\vec{B} = \mu \vec{H}$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho_f}{\epsilon} \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \vec{\nabla} \times \vec{B} = \mu \sigma \vec{E} + \mu \epsilon \frac{\partial \vec{E}}{\partial t}$$

$$\frac{\partial \rho_f}{\partial t} = -\vec{\nabla} \cdot \vec{J}_f = -\vec{\nabla} \cdot (\sigma \vec{E}) = -\sigma \vec{\nabla} \cdot \vec{E} = -\frac{\sigma}{\epsilon} \rho_f$$

$$\frac{\partial \rho_f}{\partial t} = -\frac{\sigma}{\epsilon} \rho_f \Rightarrow \rho_f = \rho_0 e^{-\frac{\sigma}{\epsilon} t} \quad \tau_f = \frac{\epsilon}{\sigma}$$

$$\rho_H = 0$$

$$\vec{\nabla} \cdot \vec{E} = 0$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}$$

$$\begin{aligned} \nabla^2 \vec{E} &= \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2} \\ \nabla^2 \vec{B} &= \mu\sigma \frac{\partial \vec{B}}{\partial t} + \mu\epsilon \frac{\partial^2 \vec{B}}{\partial t^2} \end{aligned}$$

$$\tilde{\omega} = \omega + i\gamma \quad \tilde{E} = E_0 e^{i(kz - \omega t)} \quad ; \quad \tilde{B} = B_0 e^{i(kz - \omega t)}$$

$$\vec{J}_f = \sigma \vec{E}$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho_f}{\epsilon} ; \quad \vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} ; \quad \vec{\nabla} \times \vec{B} = \mu \sigma \vec{E} + \mu \epsilon \frac{\partial \vec{E}}{\partial t}$$

$$\left. \begin{aligned} \nabla^2 \vec{E} &= \mu \sigma \frac{\partial \vec{E}}{\partial t} + \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2} \\ \nabla^2 \vec{B} &= \mu \sigma \frac{\partial \vec{B}}{\partial t} + \mu \epsilon \frac{\partial^2 \vec{B}}{\partial t^2} \end{aligned} \right\} \begin{aligned} \vec{E} &= \vec{E}_0 e^{i(kz - \omega t)} \\ \vec{B} &= \vec{B}_0 e^{i(kz - \omega t)} \end{aligned}$$

$$-k^2 \vec{E} = -i\mu\sigma\omega \vec{E} - \omega^2 \mu\epsilon \vec{E} \Rightarrow \boxed{k^2 = \tilde{\omega} \mu \epsilon + i\mu\sigma}$$

$$\underline{\tilde{k}} = k + i\kappa = \kappa e^{i\phi} \quad \underline{\tilde{k}}^2 = \tilde{\omega}\mu\epsilon + i\mu\sigma\omega \quad v = \frac{1}{\sqrt{\mu\epsilon}}$$

$$\underline{\tilde{E}} = \underline{\tilde{E}}_0 e^{-\kappa z} e^{i(kz - \omega t)}$$

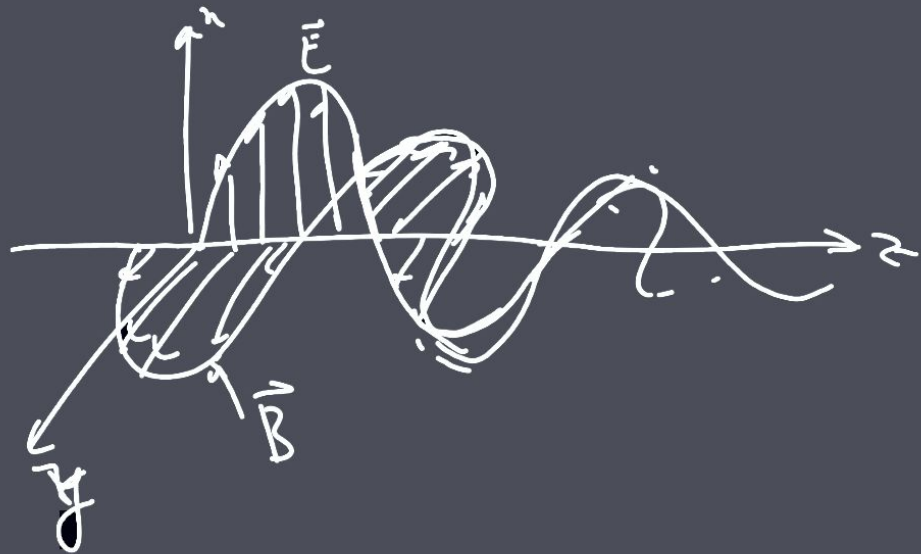
$$\vec{\nabla} \underline{\tilde{E}} = \underline{\tilde{k}}$$

$$\underline{\tilde{B}}_0 = \frac{k}{\omega} \quad \left(\phi = \frac{1}{\kappa} \right) \Leftarrow \text{Skin depth.} \quad \underline{\tilde{E}} = \underline{\tilde{E}}_0 e^{i(kz - \omega t)}$$

$$\underline{\tilde{B}} = \left(\frac{k}{\omega} \underline{\tilde{E}}_0 \right) e^{-\kappa z} e^{i(kz - \omega t)}$$

$$\underline{\tilde{E}} = E_0 \cos(kz - \omega t + \delta) \hat{n}$$

$$\underline{\tilde{B}} = \frac{E_0}{v} \cos(kz - \omega t + \delta + \phi) \hat{y}$$



Dispersion:



$$\mu_1 = \mu_2 \approx \mu_0$$

Phase velocity $v = \frac{\omega}{k}$

$$n = \sqrt{\epsilon_r} \quad n(\lambda)$$

Group velocity $v_g = \frac{d\omega}{dk}$

$$n: A + \frac{B}{\lambda^2}$$

Restoring force: $-kx$

$$F = qE = qE_0 \cos(kz - \omega t)$$

Damping: $-\gamma \frac{dx}{dt}$
 Eqn. of motion:

$$m \frac{d^2 x}{dt^2} + m\gamma \frac{dx}{dt} + m\omega_0^2 x = qE_0 \cos(kz - \omega t)$$

$$\frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = \frac{qE_0}{m} \cos(kz - \omega t)$$

$$\Rightarrow \frac{d^2 \tilde{x}}{dt^2} + \gamma \frac{d\tilde{x}}{dt} + \omega_0^2 \tilde{x} = \frac{qE_0}{m} e^{i(kz - \omega t)}$$

$$\underline{x}(t) = \frac{q E_0 / m}{(-\tilde{\omega} + \omega_0) - i\gamma\omega} e^{i(kz - \omega t)} \quad (n \sim \sqrt{\epsilon_r})$$

$$\underline{P} = q \underline{x} = \left[\frac{q^2 E_0 / m}{(-\tilde{\omega} + \omega_0) - i\gamma\omega} \right] e^{i(kz - \omega t)}$$

f_j : no. of electrons with freq ω_j & damping γ_j

N : electron/volume.

$$\vec{P} = \frac{Nq^2}{m} \sum_j \frac{f_j}{(-\tilde{\omega} + \nu_j) - i\gamma_j \omega} \vec{E}$$

$$= \epsilon_0 \chi_e \vec{E}$$

$$\begin{aligned} \vec{D} &= \epsilon_0 \vec{E} + \vec{P} \\ &= \epsilon_0 \vec{E} + \epsilon_0 \chi_e \vec{E} \\ &= \epsilon_0 (1 + \chi_e) \vec{E} \\ &= \epsilon \vec{E} \end{aligned}$$

$$\epsilon = \epsilon_0 (1 + \chi_e)$$

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} = 1 + \chi_e = 1 + \frac{Nq^2}{m\epsilon_0} \sum_j \frac{f_j}{(-\tilde{\omega} + \nu_j) - i\gamma_j \omega}$$

$$\nabla^2 \vec{E} = \mu_0 \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$v = \frac{1}{\sqrt{\mu_0 \epsilon}} = \frac{c}{n}$$

$$\vec{E} = \vec{E}_0 e^{i(kz - \omega t)}$$

$$\omega = v k \Rightarrow k = \frac{\omega}{v}$$

$$= \vec{E}_0 e^{-\kappa_1 z} e^{i(k_1 z - \omega t)}$$

$$k = \omega \sqrt{\mu_0 \epsilon} = \frac{\omega}{c} \sqrt{\epsilon_r}$$

$$= k_1 + i\kappa_1$$

$$I \propto |\vec{E}|^2$$

$\Downarrow$

$$\alpha = 2\kappa_1 \Leftarrow \text{Absorption coefficient}$$

$$\frac{1}{\epsilon} = \frac{\omega}{c} \sqrt{\epsilon_r}$$

$$\sqrt{1+\epsilon} \approx 1 + \frac{1}{2}\epsilon$$

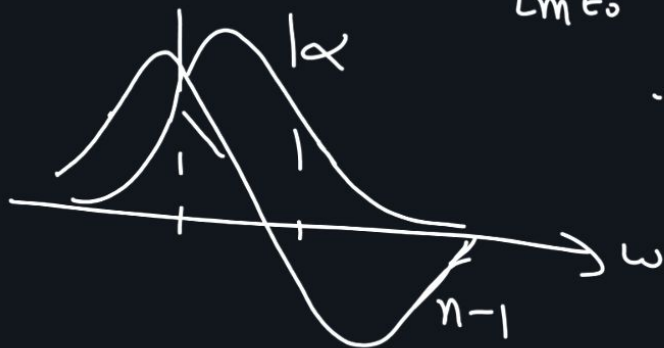
$$= \frac{\omega}{c} \left[1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{(-\tilde{\omega} + \omega_j) - i\gamma_j\omega} \right]$$

$$\epsilon_r = 1 + \frac{Nq^2}{m\epsilon_0} \sum_j \frac{f_j}{(-\tilde{\omega} + \omega_j) - i\gamma_j\omega}$$

$$= \frac{\omega}{c} \left[\underbrace{1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j (-\tilde{\omega} + \omega_j)}{(-\tilde{\omega} + \omega_j) + (i\gamma_j\omega)}}_{K_1} + i \underbrace{\frac{Nq^2\omega}{2m\epsilon_0} \sum_j \frac{f_j \gamma_j}{(-\tilde{\omega} + \omega_j) + (i\gamma_j\omega)}}_{K_2} \right]$$

$$\alpha = \frac{Nq^2}{m\epsilon_0} \sum_j \frac{\gamma_j f_j}{(-\tilde{\omega} + \omega_j) + i(\gamma_j + \omega)}$$

$$n(\omega) = \frac{c k_1}{\omega} = 1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j (-\tilde{\omega} + \omega_j)}{(-\tilde{\omega} + \omega_j) + i(\gamma_j + \omega)} \rightarrow$$



"Anomalous dispersion"

Away from resonance $\rightarrow$ forget damping.

$$\eta(\omega) \approx 1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{-\tilde{\omega} + \omega_j}$$

$$\eta(\lambda) = 1 + A \left(1 + \frac{B}{\lambda} \right) \Rightarrow (\text{Cauchy's})$$

form.

$$\frac{1}{-\tilde{\omega} + \omega_j} = \frac{1}{\omega_j} \left(1 - \frac{\tilde{\omega}}{\omega_j} \right)^{-1} \approx \frac{1}{\omega_j} \left(1 + \frac{\tilde{\omega}}{\omega_j} \right)$$

$$\eta(\omega) = 1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{\omega_j} + \frac{Nq^2 \tilde{\omega}}{2m\epsilon_0} \sum_j \frac{f_j}{\omega_j^2} = \frac{1}{\omega_j} + \frac{\omega}{\omega_j^2}$$